

■ Listable

`Listable` is an attribute that can be assigned to a symbol f to indicate that the function f should automatically be threaded over lists that appear as its arguments.

`Listable` functions are effectively applied separately to each element in a list. ■ Most built-in mathematical functions are `Listable`. ■ Example: `Log` is `Listable`. $\text{Log}[\{a, b, c\}] \longrightarrow \{\text{Log}[a], \text{Log}[b], \text{Log}[c]\}$. ■ All the arguments which are lists in a `Listable` function must be of the same length. ■ Arguments that are not lists are copied as many times as there are elements in the lists. ■ Example: `Plus` is `Listable`. $\{a, b, c\} + x \longrightarrow \{a + x, b + x, c + x\}$. ■ See page 214. ■ See also: `Thread`, `Map`.